Gavin J. Qu

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Education

University of Illinois (UIUC) | 2025-Present **Math Preparation**
MATH 416: Abstract Linear Algebra (A-); MATH 314: Intro to Proof (A-); MATH 447: Real Analysis (in-progress).

University of Essex | November 2024 ***MSc in Economics***
Honors: *Distinction (~4.0 GPA)*
Relevant coursework: Public Policy Evaluation (*Angrist & Pischke*), Data Science Economics (*ISLP/ISLR*), Graduate Econometric Methods (*Wooldridge*), Mathematical Methods (*Simon & Blume*), Advanced Microeconomic Theory (*Varian*), Behavioral Economics, Advanced Macroeconomic Theory (*Romer*).

University of Washington | June 2022 ***BA in Economics & BA in Philosophy***
Honors: Dean's List (2020-2022)
Relevant coursework: Calculus I-III, Differential Equations, Statistics, Market Design, Econometrics, Public Economics, Environmental Economics, Intermediate Micro/Macro, Industrial Organization.

Certificates: Data Structures & Algorithms; Object-Oriented Programming in Java.

Research and Projects

MSc Thesis | University of Essex | Advisor: David Zentler-Munro | 2024
"From Medical Charts to Market Forces: How Health Types Shape Labour Market Outcomes"

- Standard labor-supply models treat health as a static measure. Using 15 waves of UKHLS panel data, I cluster frailty trajectories over ages 52–60 (k-means) into latent health types following Borella et al. (2024), and test whether they predict post-60 outcomes beyond a static frailty measure. Four types emerge: resilient (61%), mild (29%), frail (9%), and a rapid-decline type (0.7%). The types reproduce some future-frailty results outside the HRS for the first time, but add little to employment, pay, or hours once contemporaneous frailty is controlled. The exception is long-term sickness and disability, where they add predictive power that survives every static benchmark and a held-out test with the clustering refit. I also show that column-mean imputation on an unbalanced panel sorts respondents by interview coverage rather than health, and correct it.

Independent Research | The Extension of MSc Thesis | Advisor: David Zentler-Munro | 2026

- Acute health shocks kill and leave survivors with labour market uncertainty. The problem I attempt to resolve is to untangle any employment effects that conflates survival with labour supply. Using 15 waves of UK panel data (UKHLS) and a baseline-healthy inception cohort aged 45–55, I estimate the labour-market cost of a first heart attack, stroke, or cancer with a staggered difference-in-differences design (Callaway–Sant'Anna) in which severity using a frailty index measured at onset, enters as a continuous dose. I estimate the following hierarchy: the shock raises mortality by 6.2 percentage points; among survivors, employment falls 11.5 points; the composite employed-and-alive total effect is −15.9 points, so 28% of the cost flows through death. Losses reflect labour-force withdrawal rather than job search, benefits and pensions replace only a third of survivors' lost labour income, and the cost is largest for the degree-educated. Honest-DiD bounds and estimator cross-checks support the design I employed.

Research Communication & Open-Source

- Creator, econ-phd-advice (open-source guide): Built and maintain a community-driven Quarto/GitHub resource hub for economics PhD and predoc applicants.
- Creator, EconJobMarket (open-source dataset & pipeline): Built a Python/SQLite ETL pipeline that scrapes and structures PhD placement outcomes from 75 top US economics programs (1987–2025) into a versioned, queryable database, with schema design distinguishing PhD-granting programs from universities and an audit trail for human-verified corrections.
- Substack & Blog (*The Causal Bit*) on Economics and Statistics.

Experience

Pre-doctoral and PhD preparation *May 2026 – Present*

Data Analyst | Costco Travel *May 2025 – May 2026*

- Conducted research-style exploratory analysis to identify drivers of operational variance; produced clear summaries for decision-making.
- Implemented statistical/econometric models (e.g., logistic regression) to quantify relationships and generate interpretable effect-based diagnostics.
- Applied machine learning methods (Random Forest, Gradient Boosting, NLP) to improve prediction and compare performance against baseline statistical models.
- Engineered reproducible Python/SQL pipelines to clean, merge, and validate large operational datasets (schema checks, missingness, outliers), improving data integrity and reducing manual work.
- Developed binary-classification models for price-variance detection (92% F1 score) with cross-validation.
- Standardized documentation and versioned workflows (LaTeX, Git) and automated ETL with Python/Bash to support scale and workflow reproducibility.
- **Technical Leadership:** Designed and hosted departmental workshops on SQL, statistical thinking, and causal inference.

Costco Corporate, Issaquah, WA *2022 — 2023*

- Conducted forecasting and EDA to identify operational inefficiencies.

Math Instructor | Juni Learning, Seattle, WA *2021 — 2022*

- Taught mathematical problem-solving and analytical reasoning from Pre-Algebra through AP Calculus.
- Assisted in designing curriculum on Economics and Personal Finance.

Technical Skills & Research Interests

Pre-doctoral and PhD preparation
Fields of Interest: Labor, Health, Education, family formation.
Methodological Interests: Causal Inference, Bayesian, and Machine Learning Methods.

Causal Inference & Econometrics: Difference-in-Differences (DiD), Regression Discontinuity Design (RDD), Instrumental Variables (2SLS/LATE), Propensity Score Matching, Randomized Controlled Trials (RCTs), Panel Data Analysis, Quantile Regression, Causal Forest, Double-debiased ML.
Machine Learning: Regularization (Lasso/Ridge), Ensemble Methods (Random Forest, Gradient Boosting), Dimensionality Reduction (PCA/PCR), Generalized Additive Models (GAMs), Classification (LDA/QDA), Neural Networks (CNN/RNN), Cross-Validation & Bootstrapping. Q-Learning.
Mathematical Methods: Dynamic Programming, Constrained & Unconstrained Optimization, Differential & Difference Equations.

Programming & Tools – Python, R, Julia, SQL, Java, Git, *LaTeX*, Bash, Tableau, Claude Code.